Machine Learning Tutorial: Predicting Titanic Survival Using Classification Models

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Github Link: https://github.com/Lahari042003/titanic-survival-ml

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1. Introduction

Machine learning is now an important tool for looking at large sets of data and making predictions based on patterns in the data. The Titanic dataset is one of the most common first problems in machine learning. It asks you to guess if a passenger survived the disaster based on their age, gender, and other factors. This problem is a binary classification task, which means that the goal is to put people into one of two groups: those who lived and those who didn't.

This report aims to illustrate the application of various machine learning techniques to this problem and assess their efficacy. There are three supervised learning models that are looked at: Logistic Regression, Decision Tree, and Random Forest. We chose these models to compare a simple linear model to more complicated non-linear and ensemble models. This report's goal is to give information about the things that helped people survive the Titanic disaster, in addition to evaluating the model.

The analysis uses a structured machine learning pipeline that includes exploratory data analysis, model training, evaluation with confusion matrices and accuracy, and interpretation of the results. The results show how machine learning can find useful patterns in old data and help people make decisions based on what they think will happen in the future.

2. Dataset Overview

There are both numbers and categories in the Titanic dataset that tell you about the people who were on the Titanic. Sex, age, passenger class, fare, number of siblings or spouses on board, number of parents or children on board, and port of embarkation are all important variables. The target variable is survival, which is shown as a binary value, with one meaning survival and zero meaning not survival.

This dataset is popular for teaching machine learning because it is both useful in the real world and not too hard to understand. It also has missing values and data types that are mixed together, which makes it a good example of how to preprocess data and test models (Kaggle, 2012).

The dataset is somewhat unbalanced because more passengers didn't survive than did. It is important to think about this imbalance because it can affect how well the model works and make predictions more likely to be correct for the majority class (He and Garcia, 2009)

3. Exploratory Data Analysis

Before using machine learning models, it is very important to do exploratory data analysis to learn about the dataset's structure and features. It helps find patterns, connections, and possible problems like missing values or skewed distributions (Han et al., 2011).

The survival distribution shows that more passengers died than lived. This imbalance suggests that a model might be able to predict non-survival with a high degree of accuracy, but such a model would not be useful in real life.

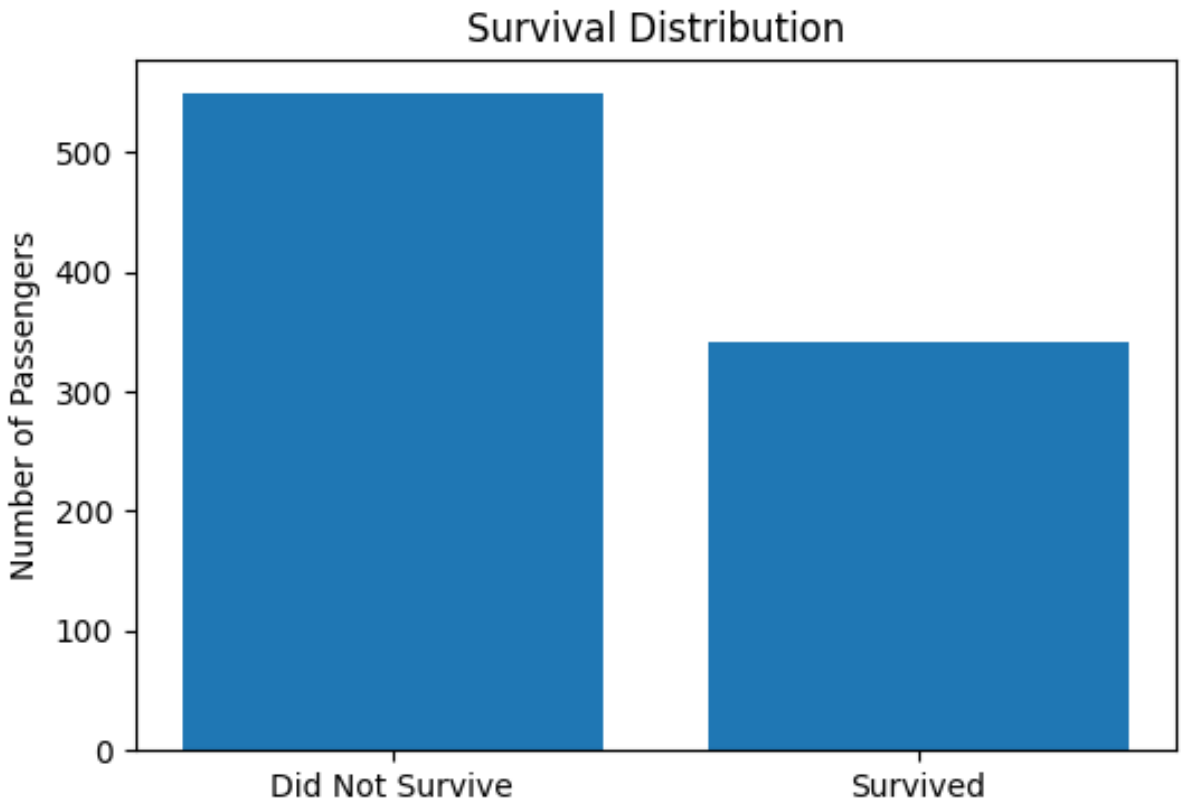

Figure 1: Distribution of survival outcomes among passengers.

Gender is an important factor in how well people survive. The analysis indicates that a significantly greater percentage of female passengers survived in comparison to male passengers. This observation corresponds with historical narratives of evacuation policies that favoured women and children. Because of this, gender is likely to be one of the most important things to look for.

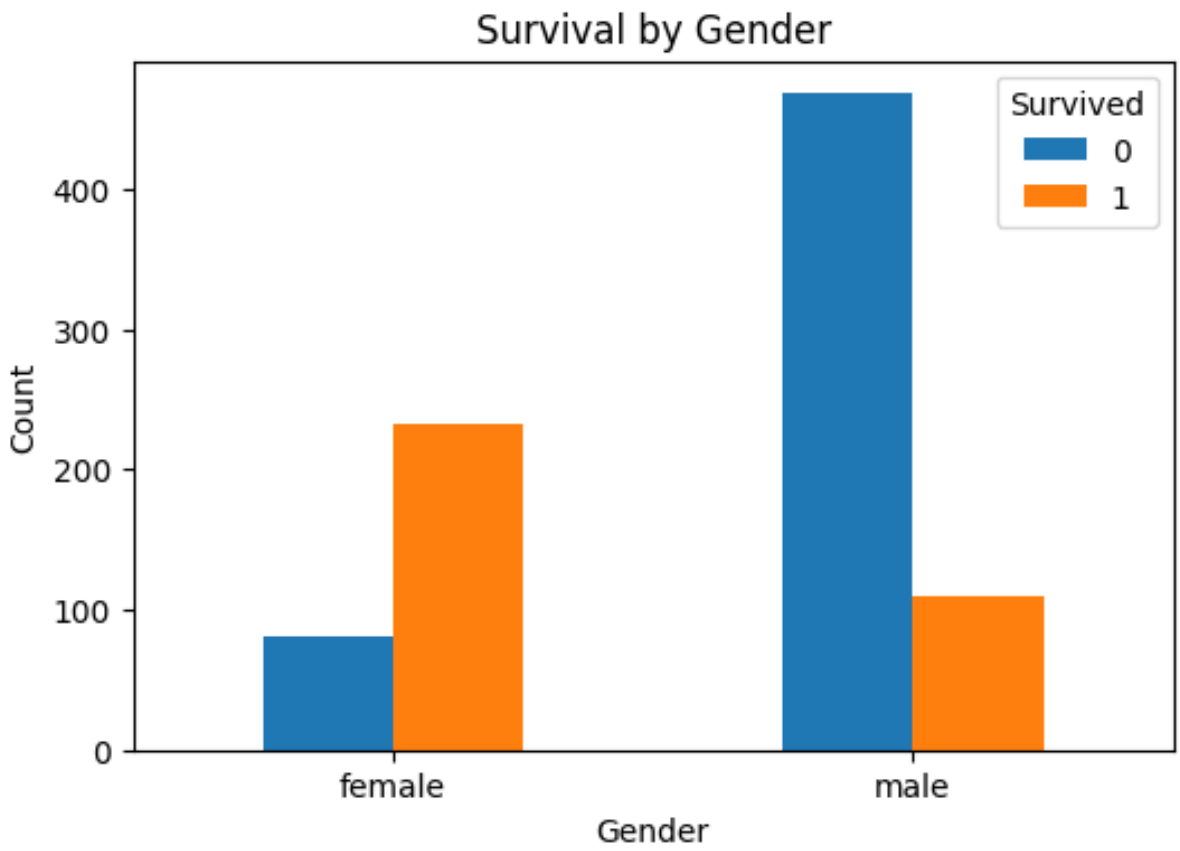

Figure 2: Survival counts grouped by gender.

The class of the passenger is also very important. The passengers in first class had the best chance of survival, while the passengers in third class had the worst. This means that socioeconomic status affected how easy it was to get to lifeboats and evacuate. During the disaster, passengers in higher classes probably had better access to information and resources.

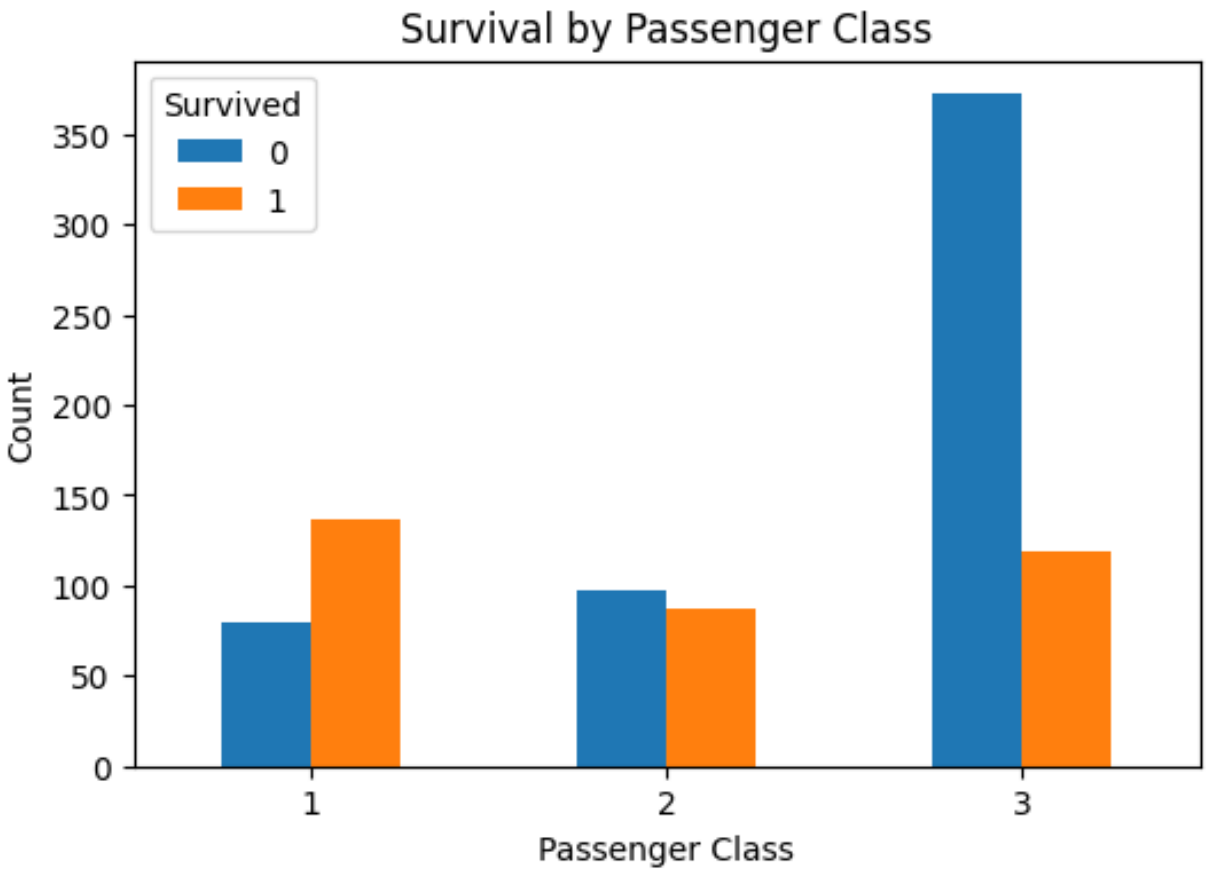

Figure 3: Survival distribution across passenger classes.

The age distribution shows that most of the passengers were between 20 and 40 years old. Even though there were fewer children, they may have had a better chance of survival because they were given priority during the evacuation. But age alone doesn't make it easy to tell who survived and who didn't, which means that it's a moderately useful feature rather than a main one.

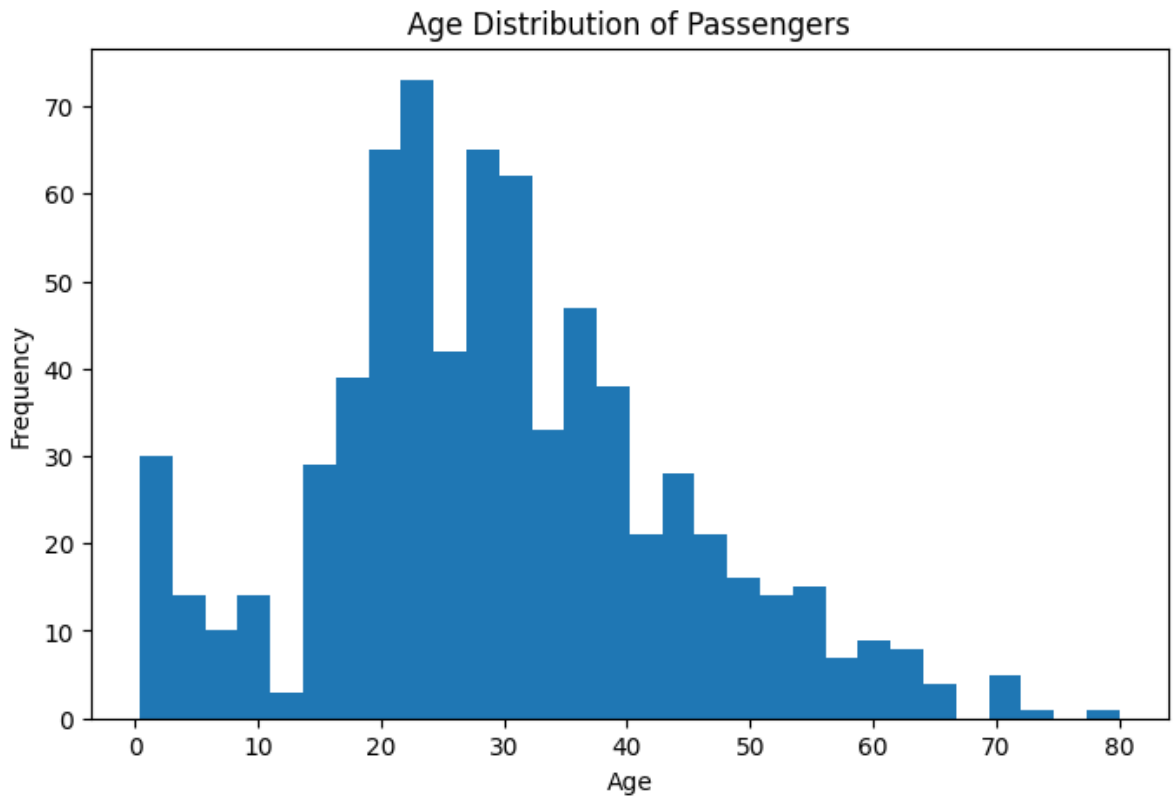

Figure 4: Distribution of passenger ages.

The fare distribution is very uneven, with most passengers paying low fares and only a few paying very high fares. Because fare is closely related to passenger class, it can be used as an indirect sign of wealth. Higher fares usually mean higher survival rates, which shows how important socioeconomic factors are.

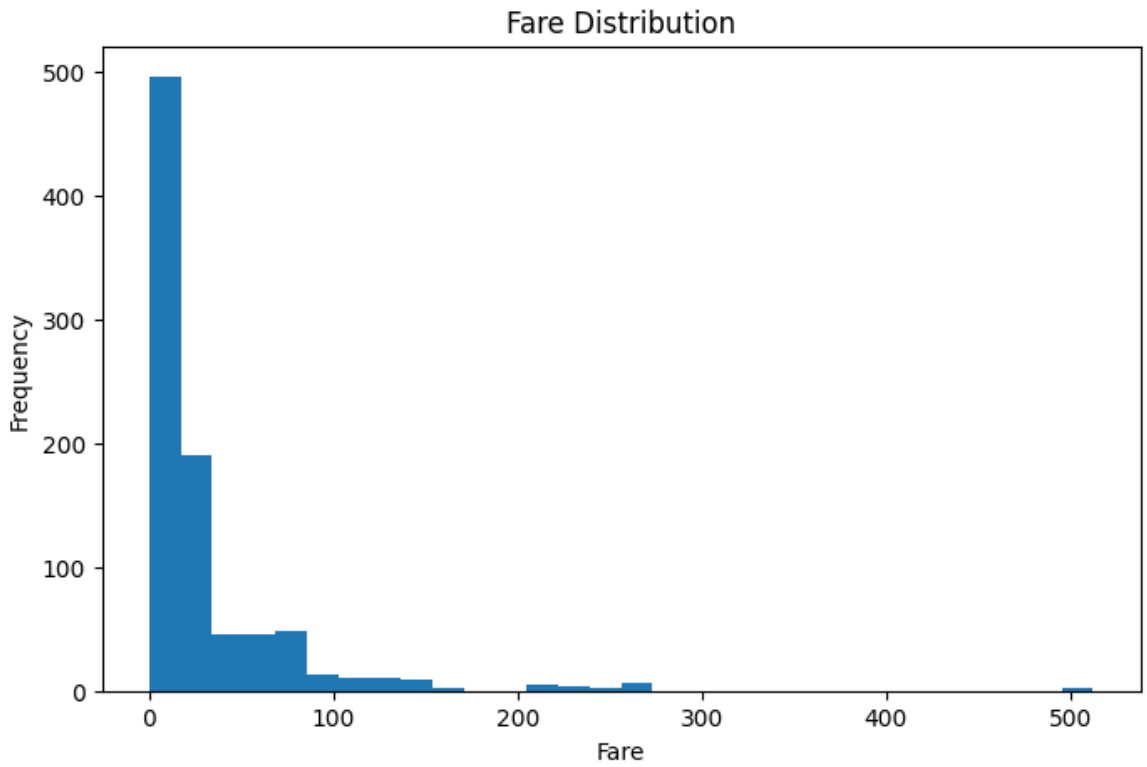

Figure 5: Distribution of passenger fares.

In general, the exploratory analysis shows that survival was not random but was affected by social and economic factors that were organised.

4. Machine Learning Models

Three classification models were used to guess how long people would live: Logistic Regression, Decision Tree, and Random Forest. These models are all different levels of complexity, which makes them useful for comparing performance and how easy they are to understand.

Logistic Regression is a linear model that people often use to solve problems with two possible answers. It uses a logistic function to figure out how likely a class is to happen, and people like it because it's easy to understand and use (Hosmer et al., 2013). The model got about 0.81% right. The confusion matrix shows that it correctly identified a lot of people who lived and a lot of people who didn't. But it did give some false negatives, which means it wrongly predicted that some passengers didn't make it when they did. Logistic Regression is a good starting point model, even though it has this one flaw.

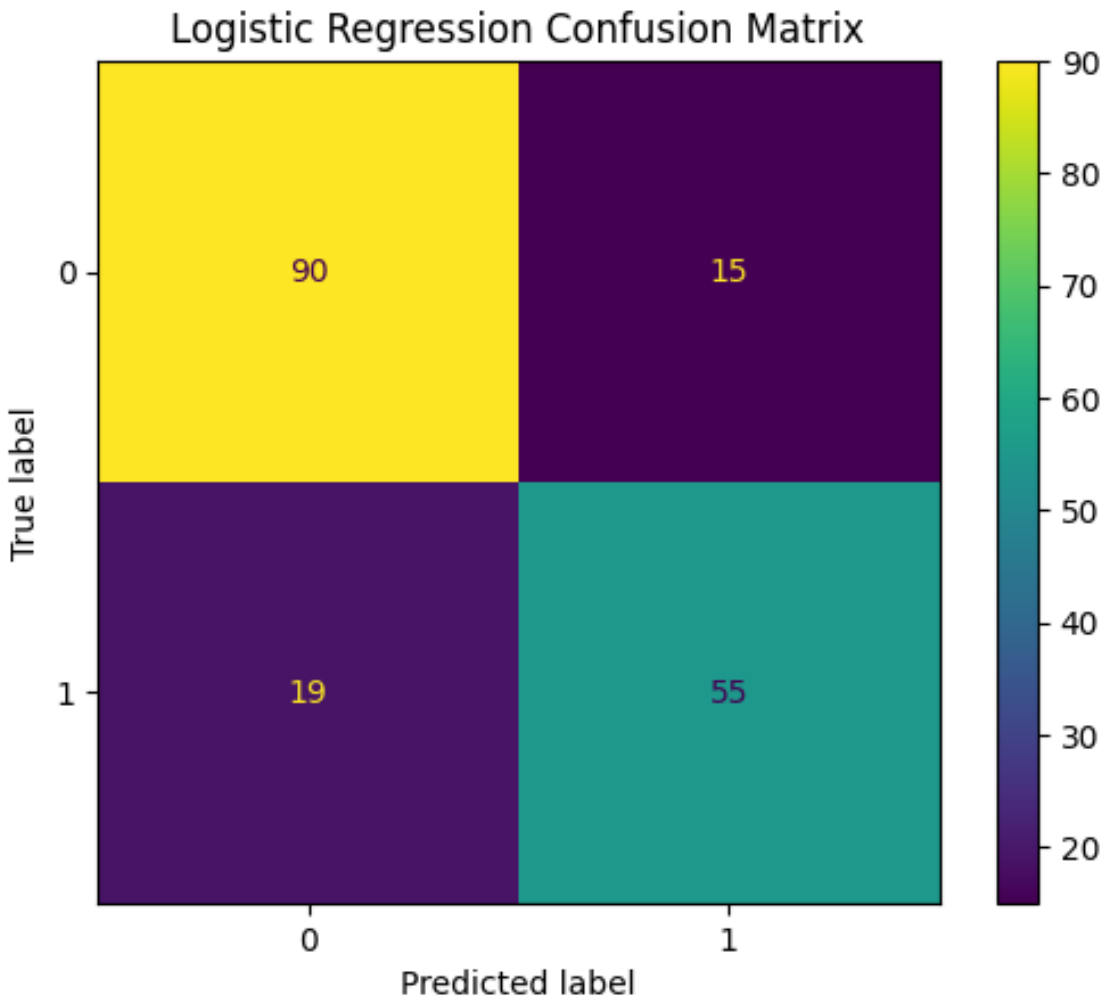

Figure 6: Confusion matrix for Logistic Regression model.

The Decision Tree model is a non-linear method that divides the data into groups based on the values of the features to make a tree-like structure of choices (Breiman et al., 1984). It got about 0.80% correct. The confusion matrix shows that it did a good job of finding people who didn't survive, but not so good at finding people who did. This is shown by the fact that there are more false negatives. It's possible that the model fit too closely to some patterns in the training data, which could make it less able to work with new data. Decision Trees are easy to understand, but they often need to be carefully tuned to work their best.

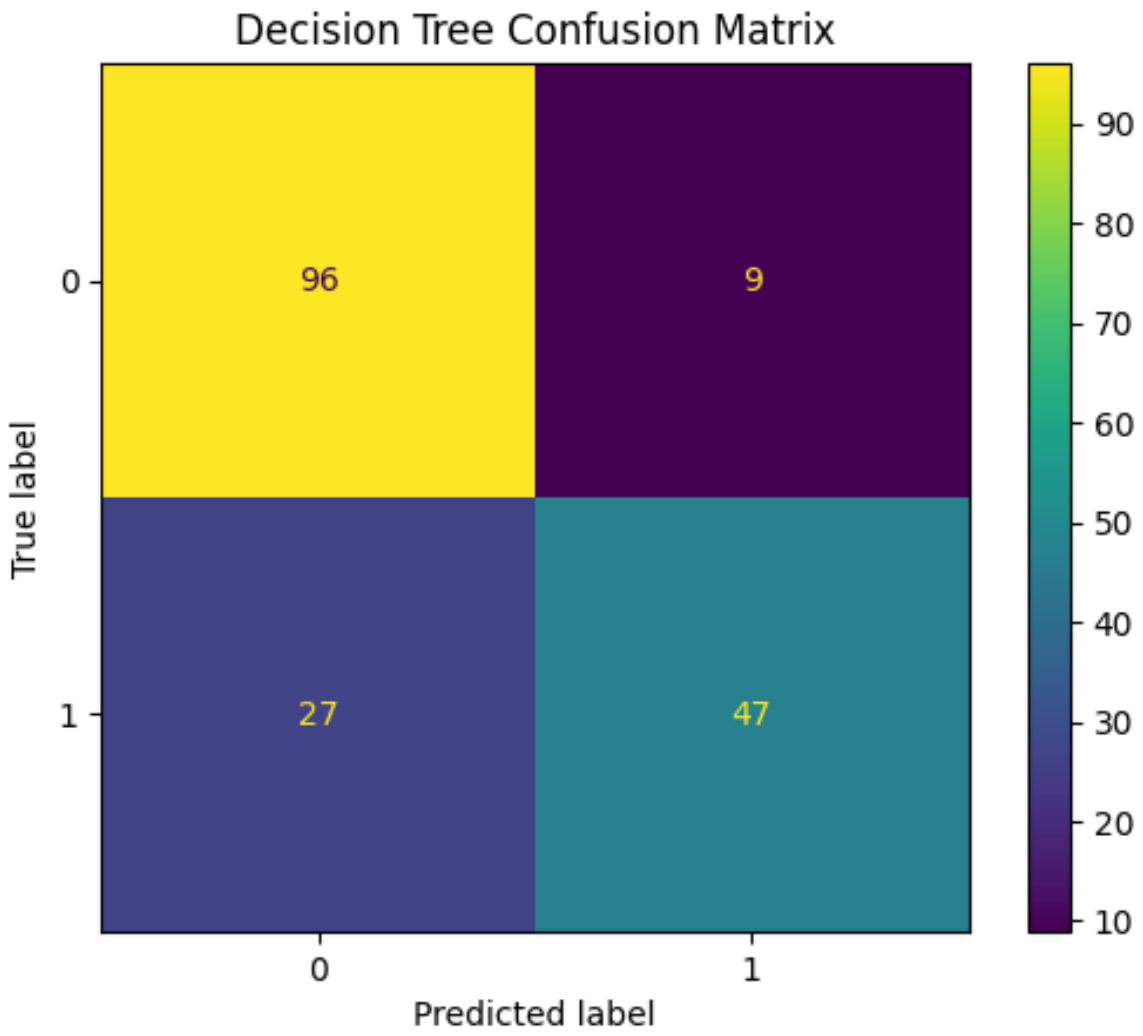

Figure 7: Confusion matrix for Decision Tree model.

Random Forest is an ensemble method that uses many decision trees to make predictions more accurate and less likely to overfit (Breiman, 2001). It got an accuracy of about 0.81, which is about the same as Logistic Regression but with a better balance between false positives and false negatives. Random Forest gets more accurate results by averaging the predictions of many trees. This lets it find more complicated relationships in the data. This makes it especially useful for datasets where features don't interact in a straight line.

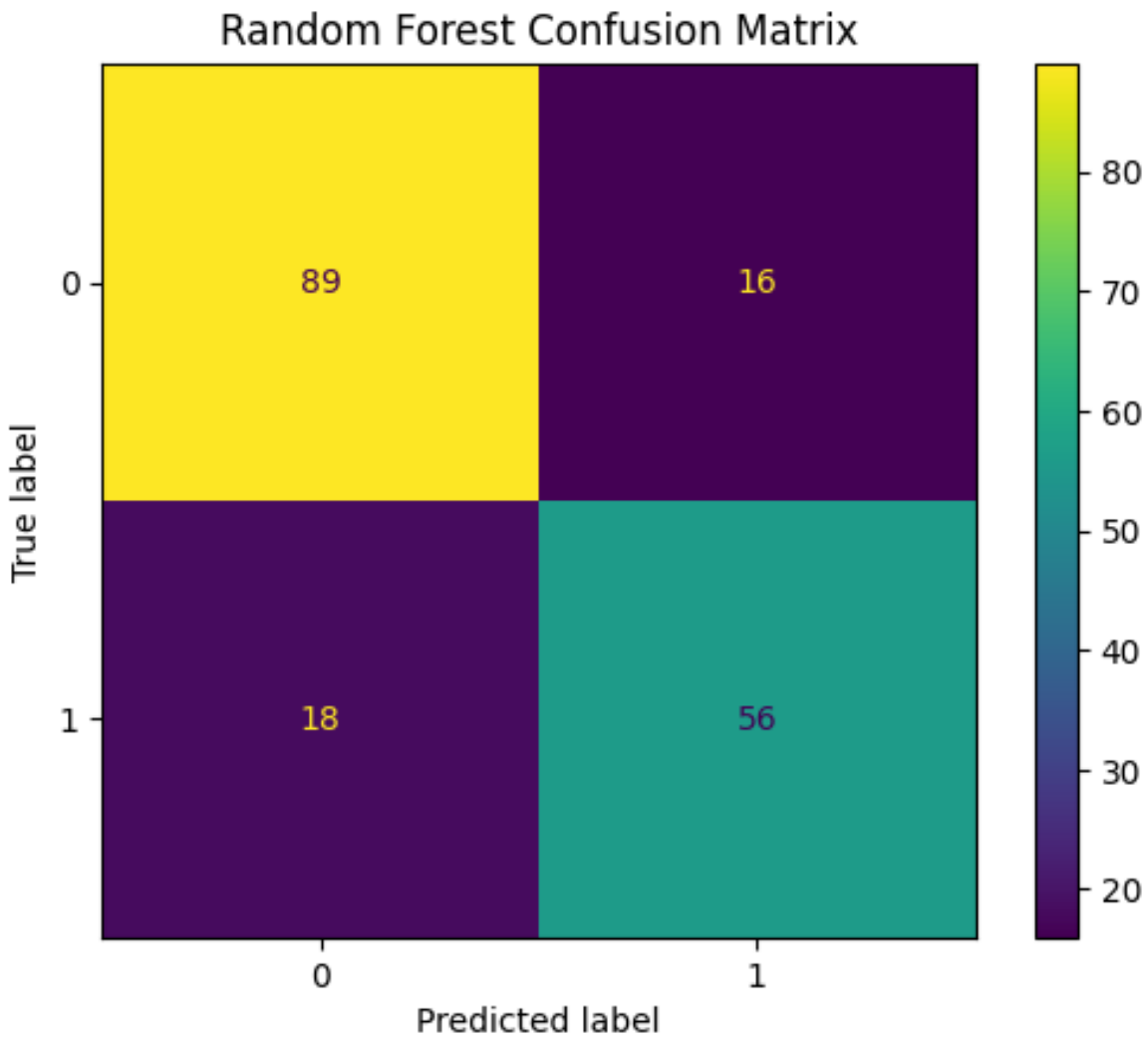

Figure 8: Confusion matrix for Random Forest model.

5. Model Comparison

When comparing the three models, it was found that Logistic Regression and Random Forest had about the same level of accuracy, while Decision Tree was a little less accurate. But accuracy alone doesn't tell the whole story about how well a model works. You should also think about how prediction errors are spread out.

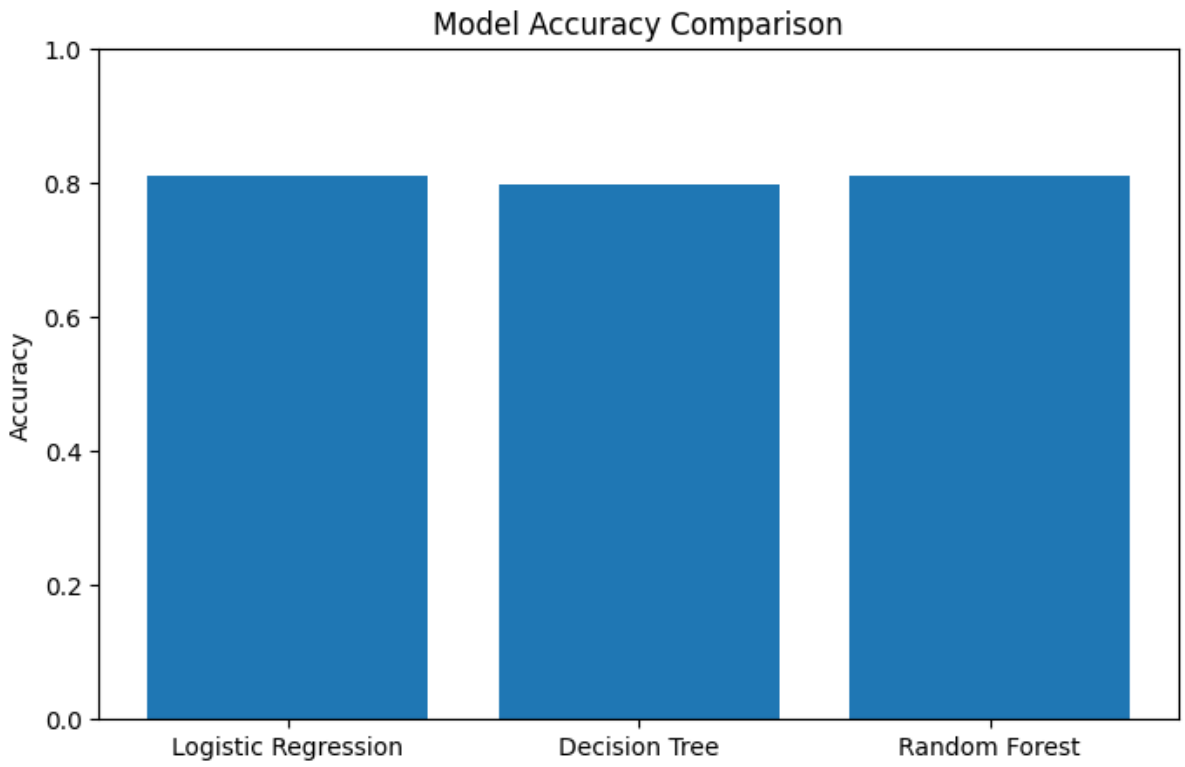

Figure 9: Accuracy comparison of machine learning models.

Logistic Regression is easy to use and understand, which makes it a good way to see how features relate to the target variable. But because it is linear, it can't pick up on patterns that are more complicated.

The Decision Tree model is easy to understand and picture, but it is easy to overfit. The results of this analysis show that it may have learned patterns that don't work well on new data.

Random Forest works best overall because it can combine different models and lower variance. It strikes a good balance between bias and variance and is better at finding non-linear relationships. Because of this, it is thought to be the best model for this dataset.

6. Feature Importance Analysis

We used the Random Forest model to do a feature importance analysis to find out which variables had the biggest effect on survival predictions. The findings indicate that fare, gender, and age are the most significant factors.

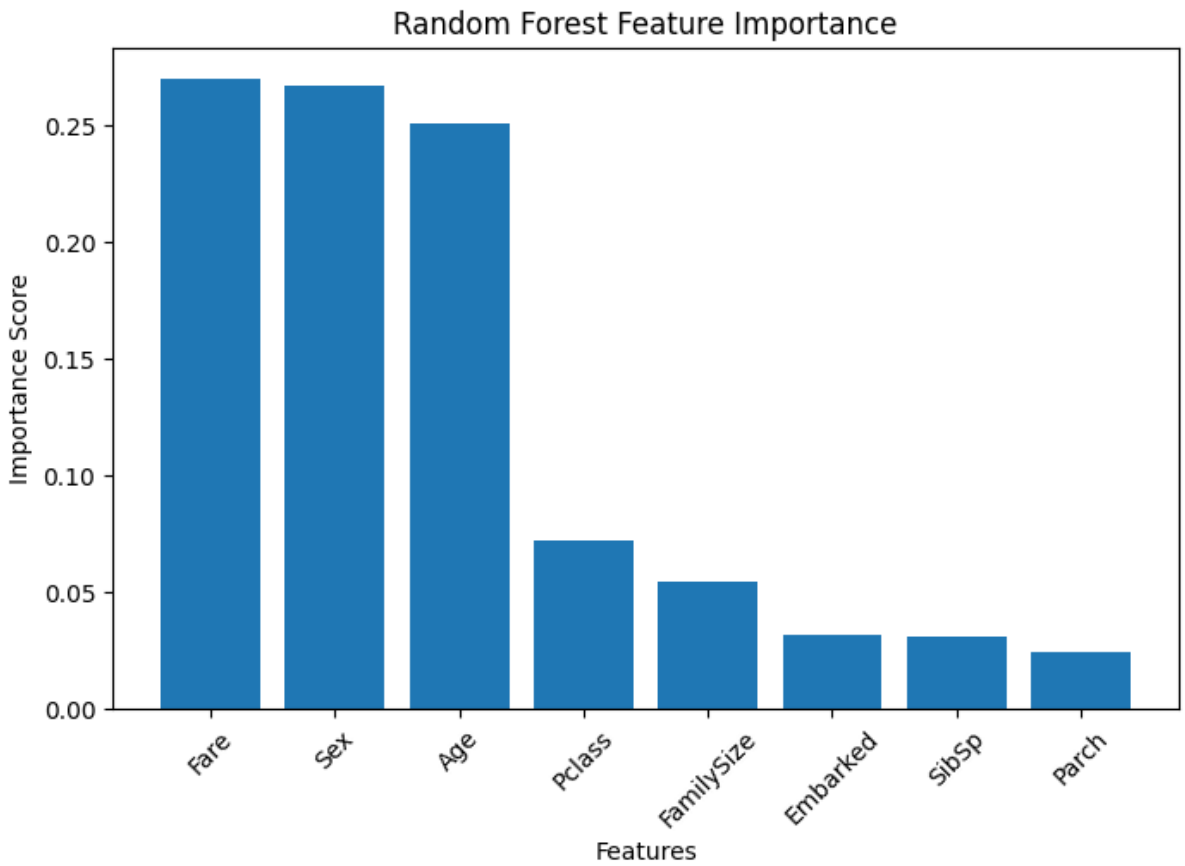

Figure 10: Feature importance scores from Random Forest model.

Fare is the most important predictor because it shows how socioeconomic status affects survival. People who paid more for their tickets were more likely to live, probably because they had better access to lifeboats and were given priority during evacuation.

Sex is also a very important factor, which shows that gender was an important factor in survival outcomes. Women who were passengers had a much better chance of living, which is in line with past evacuation policies.

Age is a part of the model, but it doesn't have as much of an effect as sex and fare. The effect of age is less clear than other factors, even though children may have had a better chance of survival.

Other factors, like the class of the passengers, the size of the family, and the port of embarkation, have lower importance scores. This doesn't mean they don't matter; it just means that their contribution to the model is smaller than that of the most important features.

7. Discussion

This analysis shows that machine learning models can find patterns in old data and give useful information. The strong impact of gender and socioeconomic status shows how organised survival was during the Titanic disaster.

The Random Forest model works best because it can model complicated relationships between variables and use ensemble learning to cut down on overfitting. Logistic Regression also works well and is a simpler option that is easy to understand. The Decision Tree model is helpful for figuring out decision rules, but it isn't as reliable without more tuning.

The fact that the models are almost the same in terms of accuracy suggests that the dataset has strong predictive signals that are not too hard to find. But the differences in error distribution show how important it is to choose a model that fits the analysis's specific goals.

8. Limitations

This analysis has some good points, but it also has some flaws that should be noted. There are missing values in the dataset, especially in the age feature. This could have an effect on how well the model works. Predictions can also be biased by the fact that there are more survivors than non-survivors.

The models also weren't fine-tuned much with hyperparameter optimisation, which could have made them work better. The analysis is also only based on one dataset, so it might not apply to other situations.

9. Future Improvements

Future research could enhance this analysis by employing cross-validation to derive more dependable performance estimates. To improve the accuracy of predictions, you could look into more advanced models like gradient boosting. Techniques for feature engineering, like pulling titles from passenger names or combining family-related variables, could give us more information.

Better handling of missing data and fixing class imbalance through resampling techniques could also make the model work better. These improvements would help make the analysis stronger and more complete.

10. Conclusion

This report has shown how machine learning can be used to predict survival outcomes using the Titanic dataset. It was shown through exploratory data analysis and model comparison that gender, fare, and age had a big effect on survival.

Random Forest was the best model overall because it could find complex relationships and avoid overfitting. Logistic Regression also did well because it was easy to use and understand.

The results show how useful machine learning is for finding patterns in data and how important it is to choose the right features and models for predictive tasks. This analysis demonstrates how data-driven methodologies can elucidate and forecast real-world results.

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